McMaster University ENGINEERING | mohawk | W BOOTH SCHOOL OF ENGINEERING PRACTICE AND TECHNOLOGY
McMaster-Mohawk Bachelor of Technology Partnership

Laboratory 2
Introduction to Tinkercad, Arduino and Sensors

SMRTTECH 3CC3

Cloud Computing and Internet of Things

B. Tech - SMRTTECH

OBJECTIVES

This lab will provide a review to important electronics concepts; and will introduce Arduino and sensors through building a simple project: an Automatic Lighting System. This system automatically turn on the light at night and turn it off in the morning.

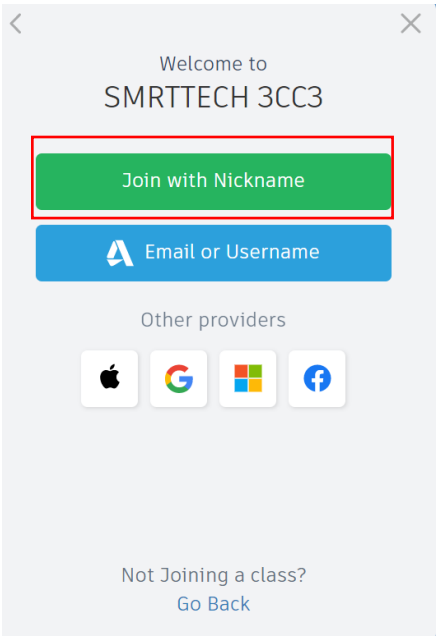
Specifically, in this lab you will do the following:

1. Connect a photoresistor (PR) to a multimeter to measure its resistance variation
2. Use a voltage divider circuit to convert resistance variations into voltage variations
3. Wire the photoresistor to an Analog Input of an Arduino
4. Wire an LED to a digital output of an Arduino
5. Write a simple code that turn on the LED when the light is below a certain threshold

PROCEDURE

EXERCISE 1 – Measure the photoresistor resistance variation with the light

1. Use this link to join the Tinkercad class.

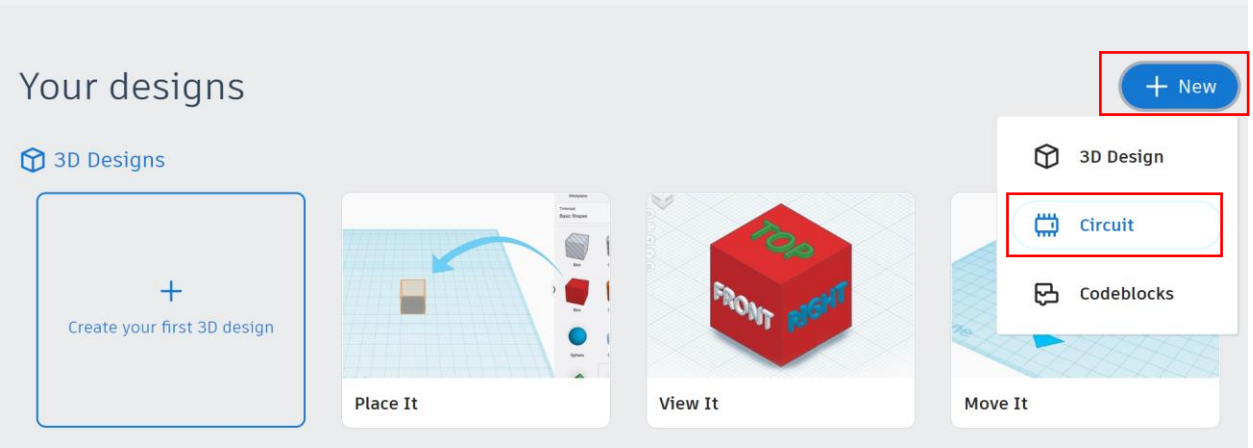


2. Click on the **Join with Nickname**.
3. Use your McMaster ID as your Nickname. This is your email address without "@mcmaster.ca" part.

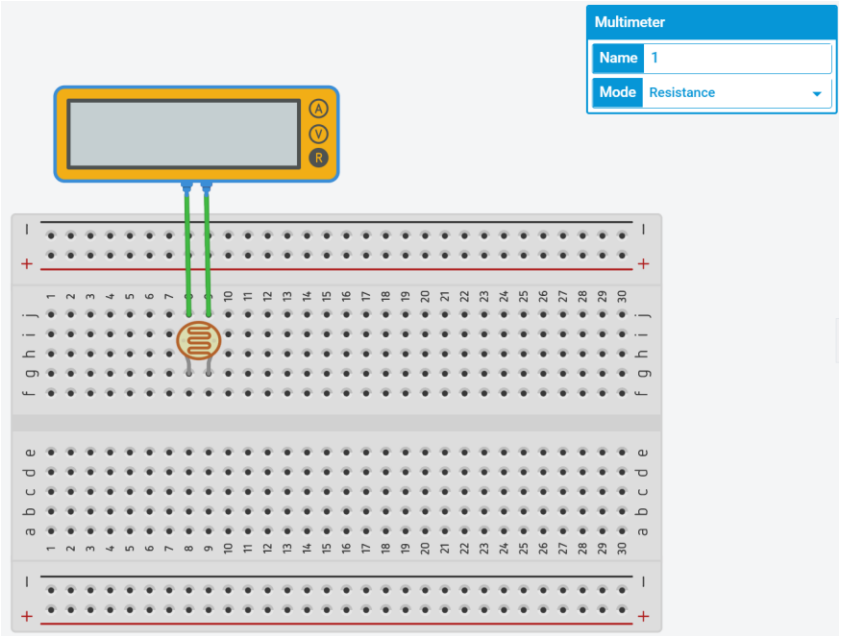
Note: if the above method did not work, you can also use your google account to access the class.

4. A Tinkercad screen should now open.

5. Select New>>Circuit.



6. Watch the video in this link and construct the circuit below.



   a. Find a photoresistor in the component list.
   b. Drag it onto the page and onto the breadboard. Both pins/legs should insert into holes on the breadboard. Place it so that the left leg is in pin G8. That means the right pin is in G9.
   c. Add a multimeter and connect its terminals to J8 and J9. Set the Multimeter mode to Resistance.

7. Use Table 1 to record the measured resistance of the PR Resistance under no light, 25% light, 50%, 75%, 100% light

Table 1

Light	0%	25%	50%	75%	100%
PR resistance					

EXERCISE 2 – Simulate a voltage divider circuit

A resistive voltage divider uses two resistors to output a voltage that is some fraction of the input voltage. The output voltage is a function directly related to the value of the two resistors. So, this circuit can allow us to convert resistance changes into voltage changes.

1. Watch the video in this link and construct the circuit below.

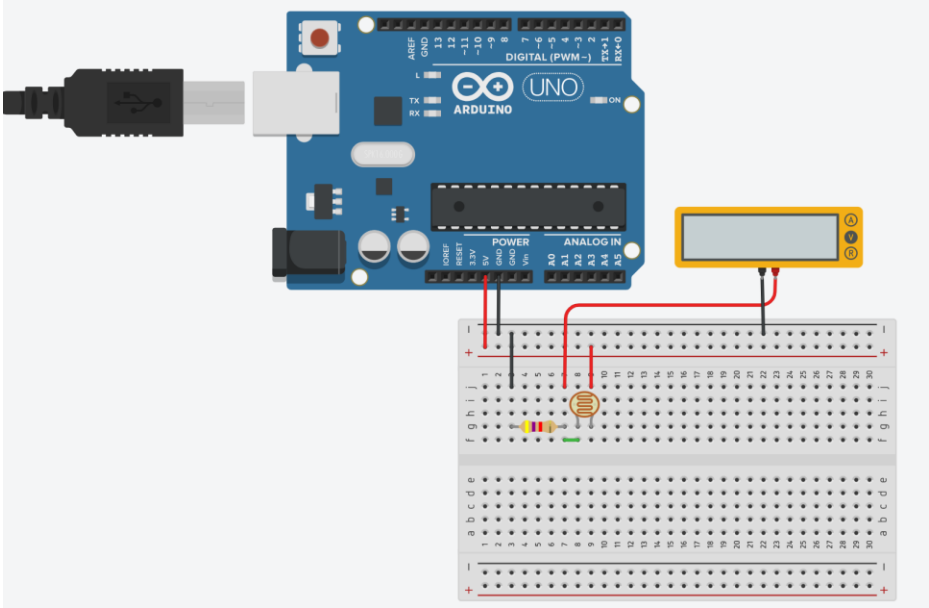


Figure 1. Voltage divider circuit

   a. Add a resistor to the circuit and place it between pins **G3** and **G7**.
      TIP: Use the rotate button in the top-left of the screen while selecting the resistor to rotate it.
   b. Set the resistor value to 4.7 KΩ.
   c. Connect pin **J3** to the **negative rail**. Connecting **J9** to the **positive rail**.
   d. Connect the negative and positive rails to the GND and 5V terminals of the Arduino.
   e. Connect the black terminal of the multimeter to the negative rail and the red terminal to **J7**. Set the multimeter mode to Voltage.

2. Use Table 2 to record the measured voltage under no light, 25% light, 50%, 75%, 100% light

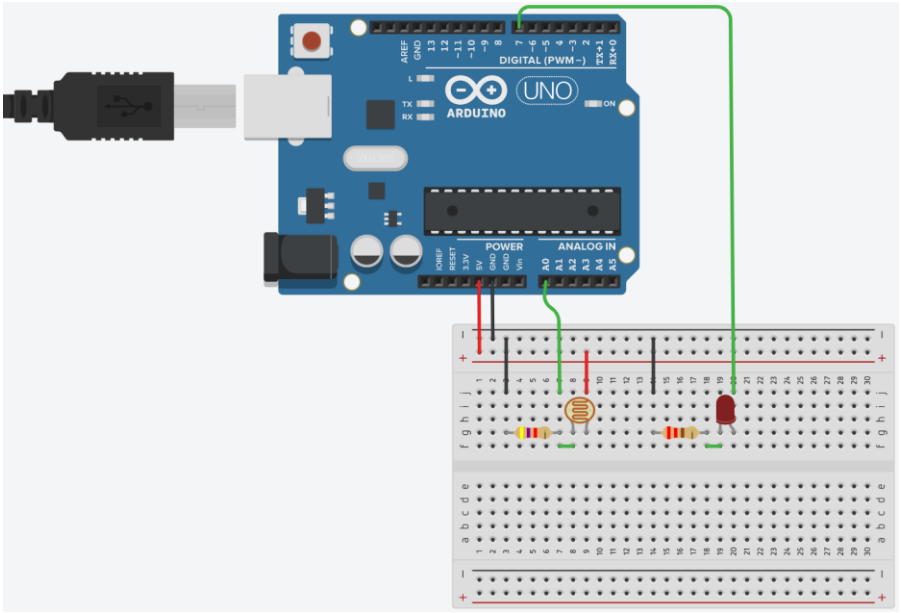
Table 2

Light	0%	25%	50%	75%	100%
Voltage					

EXERCISE 3 – Connect the Photoresistor and the LED to an Arduino

Your goal is to connect the voltage of the voltage divider circuit of EXERCISE 2 to the Arduino Analog Input (**A0**) and to connect an LED to Arduino Digital Output (**7**).

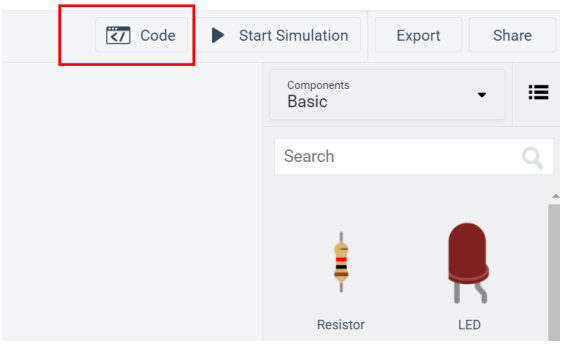
1. Construct the circuit below.



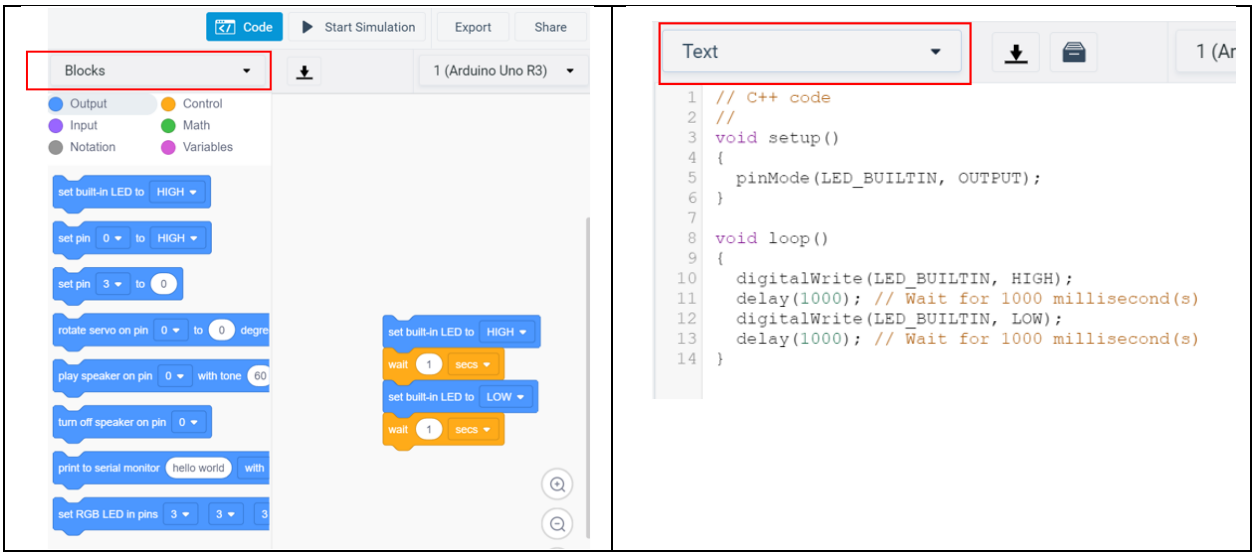
   a. Delete the multimeter.
   b. Connect **J7** to Analog Input **A0** of the Arduino.
   c. Add an LED at pins **G19** and **G20** of the breadboard.
   d. Add a resistor at pins **G14** and **G18** of the breadboard.
   e. Set the resistor value to 220 Ω.
   f. Connect pins **F18** and **F19** using a wire.
   g. Connect **J14** to the **GND** and **J20** to Digital Output pin (**7**) of the Arduino.

EXERCISE 4 – Program the Automatic Lighting System

1. Click on the Code button at the top right corner of Tinkercad



2. Change the edit mode form Blocks to Text



3. Change the program to the following.

```
int PRPin = 0;
int PR = 0;
int LEDPin = 7;
int lowLightLim = 400;

void setup()
{
  pinMode(LEDPin, OUTPUT);
}

void loop()
{
  PR = analogRead(PRPin);
  if (PR<lowLightLim){
    digitalWrite(LEDPin, HIGH);
  }
```

```
  else{
    digitalWrite(LEDPin, LOW);
  }
}
```

EXERCISE 5 – Construct a voltage divider circuit with a real photoresistor

Equipment required

- Trainer board with internal power supply - Quantity 1
- Photoresistor - Quantity 1
- 4.7 kΩ resistor (0.25 Watt) - Quantity 1
- Multimeter - Quantity 1

1. Use the above equipment to construct the circuit in Figure 1. Use the trainer board 5 V supply instead of the Arduino 5V terminals.
2. Use a multimeter to measure the total resistance of the circuit under the following circumstances:
    a. Dark condition (use your finger to totally cover the top surface of the sensor)
    b. Normal Light
    c. Flashlight (use your phone flashlight on top of the surface of the sensor).
3. Use Table 3 to record the measurements in step 2.

Table 3

Light	Dark	Normal	Flashlight
Total Resistance (kΩ)			

4. Use the voltage divider rule formulae to calculate the theoretical sensor voltage under dark, normal, and flashlight conditions. Use Table 4 to tabulate your results.
5. Use a voltmeter as shown in figure 1 to measure the sensor voltage under dark, normal, and flashlight conditions. Use Table 4 to tabulate your results.

Table 4

Light	Dark	Normal	Flashlight
Voltage (V) [Calculated]			
Voltage (V) [Measured]			

SUBMISSION

1. There is no report required for this lab.
2. You have to show the results of Tables 1-4 to the lab instructor.
3. You have to demonstrate the functionality of your circuit in EXERCISE 4 to the lab instructor.
4. You have to demonstrate the functionality of your circuit in EXERCISE 5 to the lab instructor.